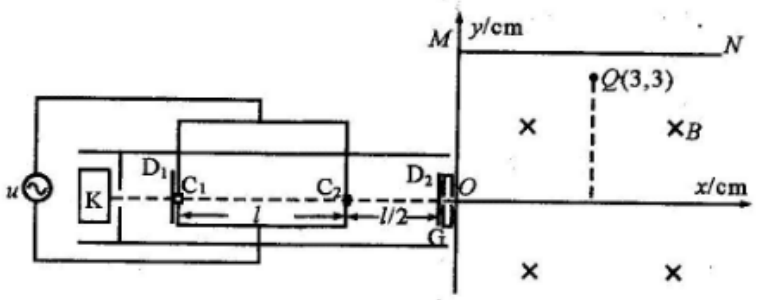
5. (25 points)

K emits positive particles continuously at various speeds, with charge $q$ and mass $m$, all along the dashed line. $D_1$、$D_2$ are timed shutters; $C_1$、$C_2$ are gates; $C_1$ adjoins $D_1$; gate spacing $l$. Both gates carry AC voltage $u=U_0\sin(\frac{2\pi}{T}t)$ of period $T$. Passage occurs only instantaneously near zero voltage. Energy modulator G adjoins $D_2$, $l/2$ from gate $C_2$. At time $t$, G changes energy by $\Delta E_t=E_0\sin(\frac{2\pi}{T}t+\frac{\pi}{4})$, keeping velocity direction. $E_0=\frac{\sqrt{2}}{4}\times m\frac{l^2}{T^2}$. Past G, particles enter at O the uniform field right of G. Its magnitude is $B$, into the page; all is in vacuum. As shown, origin O defines Cartesian axes $Oxy$; MN is parallel to the $x$ axis. At $t=0$ open $D_1$, $D_2$ together to admit particles to $C_1$. At $t=3T/4$, close $D_1$ to stop entry to $C_1$; at $t=10T/4$, close $D_2$ to stop entry to G. The fastest entry at O passes Q at (3cm, 3cm). For the slowest entry at O to pass Q, where must boundary MN be shifted parallel to itself in plane $Oxy$?

**5. Source K shown emits particles continuously at various speeds, with positive charge $q$ and mass**




**$m$, all along the dashed line. Timed shutters $D_1$、$D_2$ open and close. $C_1$、$C_2$ are gates; $C_1$ adjoins $D_1$, spacing $l$. Both gates carry AC voltage $u = U_0\sin\left(\frac{2\pi}{T}t\right)$, of period $T$. Particles pass only for an instant near zero gate voltage. Modulator G adjoins $D_2$; distance to gate $C_2$ is $l/2$. At time $t$, passing G changes the energy by $\Delta E_t = E_0\sin\left(\frac{2\pi}{T}t + \frac{\pi}{4}\right)$; direction unchanged. $E_0 = \frac{\sqrt{2}}{4}\times$**